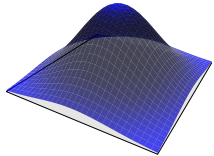
TECHNISCHE UNIVERSITÄT WIEN



Lecture 3

Statistics and Probability Theory

Discrete Probability Spaces and Fundamentals

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Probability Space

Random Experiment

is a *repeatable* procedure with *well-defined possible outcomes*.

Sample Space

is the *set of all possible outcomes* with $\omega \in \Omega$ *are the elementary events*.

Events

a *subset of the sample space*, that is $A \subseteq \Omega$.

Often described in natural language, for example
$\emptyset$ = impossible event, Ω = sure event,
A^c = opposite of event A

Set of all Events

$\mathfrak{S} \subseteq 2^\Omega$ is a σ-Algebra.

Contains $\Omega \in \mathfrak{S}$ and is closed under complement, countable unions and intersections.

Probability measure

assigns probabilities to events.

$P : \mathfrak{S} \to [0, 1]$

Axioms

$P(\Omega) = 1$, $P(A) \geq 0$ for all $A \in \mathfrak{S}$,
$P(A \dot\cup B) = P(A) + P(B)$ for all disjoint $A, B \in \mathfrak{S}$

Probability Space

$(\Omega, \mathfrak{S}, P)$ is called a *Probability Space*

Discrete Probability Spaces

Countable number of elementary events

A *discrete probability space* is determined by the *Sample Space* being *countable*.

$$\Omega = \{\omega_1, \omega_2, \omega_3, \ldots\}$$

The *probability* mass function (in short p.m.f.) $P : \mathfrak{S} \to [0, 1]$ is *uniquely defined* by the *probability of the elementary events*

$$p_i = P(\{\omega_i\})$$

Probability tables provide mappings;

events ω_i	ω_1	ω_2	$\ldots$	ω_i	$\ldots$
probability p_i	p_1	p_2	$\ldots$	p_i	$\ldots$

where

$$0 \leq p_i \leq 1 \text{ and } \sum_i p_i = 1$$

The probability of an event $A \subseteq \Omega$ is

$$P(A) = \sum_{\omega \in A} P(\{\omega\}) = \sum_{\omega_i \in A} p_i$$

Discrete Probability Spaces

We toss a *fair coin* twice.

RANDOM EXPERIMENT: toss the fair coin twice and list the outcome.

SAMPLE SPACE: $\Omega = \{HH, HT, TH, TT\} = \{\omega_1, \omega_2, \omega_3, \omega_4\}$.

σ-ALGEBRA: Is the power set (set of all subsets) of size $|\mathfrak{S}| = 2^{|\Omega|} = 2^4 = 16$

$$\mathfrak{S} = 2^{\Omega} = \{\emptyset, \{HH\}, \{HT\}, \{TH\}, \{TT\}, \{HH, HT\}, \{HH, TH\}, \{HH, TT\}, \{HT, TH\}, \{HT, TT\}, \{TH, TT\}, \{HH, HT, TH\}, \{HH, HT, TT\}, \{HH, TH, TT\}, \{HT, TH, TT\}, \Omega\}$$

PROBABILITY: All the outcomes are equally likely (fair coin)

ω	HH	HT	TH	TT
$P(\omega)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

EVENTS: Examples:

One head in two tosses

$$A = \{HT, TH\}$$

$$P(A) = P(\{HT\}) + P(\{TH\}) = \frac{1}{2}$$

At least one head

$$B = \{HH, HT, TH\} = \Omega \setminus \{TT\}$$

$$P(B) = 1 - P(\{TT\}) = 1 - \frac{1}{4} = \frac{3}{4}$$

The *sum of two 6-sided dice*;

Random Experiment: Roll two 6-sided dice and take the sum.

Version 1

Sample Space: Dice rolls as elementary events (tuples of number from 1 to 6)
$\Omega = \{(1,1),(1,2),(1,3),\ldots,(6,5),(6,6)\}$.

Probability: All outcomes are equally likely (using fair dice)

$$P(\{(x,y)\}) = \frac{1}{36}, \qquad x,y \in \{1,2,3,4,5,6\}$$

Events: The event $S_{10} = [\text{sum is } 10] = \{(4,6),(5,5),(6,4)\} \subseteq \Omega$ has probability

$$P(S_{10}) = \frac{|S_{10}|}{|\Omega|} = \frac{3}{36} = \frac{1}{12}$$

The *sum of two 6-sided dice*;

RANDOM EXPERIMENT: Roll two 6-sided dice and take the sum.

VERSION 2

SAMPLE SPACE: The sum of dice $\Omega = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$.

PROBABILITY: The possible sums are *not* equally likely (using fair dice)

$$P(\{\omega\}) = \frac{\min(\omega - 1, 13 - \omega)}{36}, \qquad \omega \in \Omega$$

(Derived by counting the number of tuples giving the same sum from Version 1)

EVENTS: The event $S_{10} = [\text{sum is } 10] = \{10\}$ has probability

$$P(S_{10}) = P(\{10\}) = \frac{\min(9, 3)}{36} = \frac{1}{12}$$

Discrete Probability Spaces

We toss a *fair coin until the first head*.

SAMPLE SPACE: Countable but infinitely many elementary events;
$\Omega = \{H, TH, TTH, TTTH, TTTTH, \ldots\}$,

PROBABILITY: The probability table is infinitely "big"

ω	H	TH	TTH	$TTTH$	$TTTTH$	$\ldots$
$P(\omega)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\ldots$

Does it sum to one? Yes, it is a Geometric series!

EVENTS A: We throw the coin at most 4 times; $A = \{H, TH, TTH, TTTH\}$

$$P(A) = P(\{H\}) + P(\{TH\}) + P(\{TTH\}) + P(\{TTTH\}) = \frac{15}{16} = 93.75\%$$

EVENTS B: We throw the coin more than 4 times; $B = A^c = \Omega \setminus A$

$$P(B) = 1 - P(A) = \frac{1}{16} = 6.25\%$$

Count the *number of taxis* passing through a street in an hour.

SAMPLE SPACE: $\Omega = \mathbb{N}_0 = \{0, 1, 2, 3, \ldots\}$

PROBABILITY: Often modelled with the p.m.f.

$$p_k = P(\{k\}) = e^{-\lambda}\frac{\lambda^k}{k!}$$

for $k \in \mathbb{N}_0$, known as the *Poisson distribution* with *parameter* $\lambda \in (0, \infty)$ describing the average number of taxis.

Does it sum to one?

$$P(\Omega) = \sum_{k=0}^{\infty} P(\{k\}) = e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = e^{-\lambda}e^{\lambda} = 1.$$

Continuous Probability Spaces

How long does a lite bulb last?

RANDOM EXPERIMENT: Take a lite bulb (all of the same making) and let it burn till it dies.

SAMPLE SPACE: $\Omega = [0, \infty) \subset \mathbb{R}$, any non-negative real value!

PROBABILITY: We now have a continuum of possible outcomes!

Therefore we need the concept of a *probability density function* (in short p.d.f.) and *replace summation with integration*.

We will talk about that next lecture.

From now on we assume $(\Omega, \mathfrak{S}, P)$ to be an arbitrary probability space

Kolmogorov Axioms

1. Non-Negativity:
$$P(A) \geq 0 \qquad \text{for all } A \in \mathfrak{S}$$
2. Unit-Measure:
$$P(\Omega) = 1$$
3. σ-Additivity: For any sequence of disjoint events A_i holds
$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Rules

Given two events A, B then
A^c, $A \cap B$ and $A \cup B$ are also events and the following holds;

(Finite) Additivity: $A \cap B = \emptyset$ implies $P(A \cup B) = P(A) + P(B)$.

First set $A_1 = \Omega$ and $A_i = \emptyset$ for all $i \geq 2$, then

$$P(\Omega) = P\left(\bigcup_{i=1}^{\infty} A_i\right) = P(\Omega) + \sum_{i=2}^{\infty} P(\emptyset)$$

which implies $P(\emptyset) = 0$.

Now set $A_1 = A, A_2 = B$ and $A_i = \emptyset$ for all $i \geq 3$, then the σ-Additivity reduces to

$$P(A \cup B) = P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) = P(A_1) + P(A_2) + \sum_{i=3}^{\infty} P(\emptyset) = P(A) + P(B).$$

Mutual Exclusivity: $A \cap B = \emptyset$ implies $P(A \cap B) = 0$.

We already know that $P(\emptyset) = 0$, therefore $P(A \cap B) = P(\emptyset) = 0$.

COMPLEMENT: $P(A^c) = 1 - P(A)$

$$1 \overset{2}{=} P(\Omega) = P(A \cup A^c) \overset{3}{=} P(A) + P(A^c)$$

INCLUSION-EXCLUSION PRINCIPLE: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$P(A \cup B) \overset{3}{=} P(A \setminus (A \cap B)) + P(B) \tag{1}$$

$$P(A) \overset{3}{=} P(A \setminus (A \cap B)) + P(A \cap B) \tag{2}$$

$$P(A \cup B) - P(A) = P(B) - P(A \cap B) \tag{1 - 2}$$

MONOTONICITY: $A \subseteq B$ implies $P(A) \leq P(B)$

$$P(A) \overset{1}{\leq} P(A) + P(B \setminus A) \overset{3}{=} P(B)$$

to list a few, Venn diagrams are useful for understanding.

Conditional Probability

Given more information changes the Probability

Conditional probability answers the question of how the *probability of an event changes when we have additional information.*

Conditional Probability

The *conditional probability* of A *given* B *if* $P(B) > 0$ is *defined* by

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Useless Information: If $B = \Omega$ then $P(A \mid B) = P(A \mid \Omega) = P(A)$

Disjoint Events: If $A \cap B = \emptyset$ then $P(A \mid B) = 0$

All Information: If $A \supseteq B$ then $P(A \mid B) = 1$

Conditional Probability

What is the probability of the *sum of two 6-sided dice being* 9, denoted S_9, *given that one of the dice is a* 5, denoted D_5?

Setting up the sample space $\Omega = \{(1,1),(1,2),\ldots,(6,6)\}$ as all tuples of 1 to 6 then

$$S_9 = \{(3,6),(4,5),(5,4),(6,3)\},$$
$$D_5 = \{(1,5),(2,5),(3,5),(4,5),(5,5),(6,5),(5,1),(5,2),(5,3),(5,4),(5,6)\}$$

leading to

$$P(S_9) = \frac{|S_9|}{|\Omega|} = \frac{1}{9}, \qquad P(D_5) = \frac{|D_5|}{|\Omega|} = \frac{11}{36}.$$

Knowing one die is a 5 changes the probability of obtaining 9 as the sum of dice

$$P(S_9 \mid D_5) = \frac{P(S_9 \cap D_5)}{P(D_5)} = \frac{2/36}{11/36} = \frac{2}{11}$$

Intuition, assuming equal probability for elementary event.

What is the probability of the *sum of two 6-sided dice being* 9, denoted S_9, *given that one of the dice is a* 5, denoted D_5?

	1	2	3	4	5	6
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

$$P(S_9 \mid D_5) = \frac{|S_9 \cap D_5|}{|D_5|} = \frac{|\{(4,5),(5,4)\}|}{|D_5|} = \frac{2}{11}$$

Independence

being unrelated

Independence

If for two events A, B holds

$$P(A \cap B) = P(A)P(B)$$

then we say *A and B are independent*.[2]

If A and B are independent (and $P(B) > 0$) then

$$P(A \mid B) \overset{\text{(cond)}}{=} \frac{P(A \cap B)}{P(B)} \overset{\text{(ind)}}{=} \frac{P(A)P(B)}{P(B)} = P(A)$$

The *reverse conclusion, is also true*, if $P(A \mid B) = P(A)$ (and $P(B) > 0$) then $A \perp\!\!\!\perp B$.

[2] often written as $A \perp\!\!\!\perp B$ or $A \perp B$

Independence

more than just two

Three events A, B, C we call *independent* if they are *pair-wise independent*

$$P(A \cap B) = P(A)P(B), \quad P(A \cap C) = P(A)P(C), \quad P(B \cap C) = P(B)P(C)$$

and

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

Independence

Throwing a *fair coin three times*, we get

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

Let A_i *be the event that the i'th toss is heads*, for example

$$A_1 = \{\boldsymbol{H}HH, \boldsymbol{H}HT, \boldsymbol{H}TH, \boldsymbol{H}TT\}, \text{ with } P(A_1) = \tfrac{1}{2}$$

then

$$P(A_i \cap A_j) = \frac{1}{4} = P(A_i)P(A_j), \qquad i, j \in \{1, 2, 3\} \text{ such that } i \neq j$$

and

$$P(A_1 \cap A_2 \cap A_3) = P(\{HHH\}) = \frac{1}{8} = P(A_1)P(A_2)P(A_3)$$

meaning that A_1, A_2 *and* A_3 *are independent*.

Independence

Throwing a *fair coin three times*, we get

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

Now let B_i be the event that the i'th toss is tails, for example

$$B_2 = \{H\boldsymbol{T}H, H\boldsymbol{T}T, T\boldsymbol{T}H, T\boldsymbol{T}T\}, \text{ with } P(B_2) = \tfrac{1}{2}$$

and C_i the event that all but the i'th throw are identical, for example

$$C_3 = \{\boldsymbol{HH}H, \boldsymbol{HH}T, \boldsymbol{TT}H, \boldsymbol{TT}T\}, \text{ with } P(C_3) = \tfrac{1}{2}$$

then A_1, B_2 and C_3 are "pair-wise independent" but *not independent*, because

$$P(A_1 \cap B_2) = P(\{HTH, HTT\}) = \tfrac{1}{4} = P(A_1)P(B_2)$$
$$P(A_1 \cap C_3) = P(\{HHH, HHT\}) = \tfrac{1}{4} = P(A_1)P(C_3)$$
$$P(B_2 \cap C_3) = P(\{TTH, TTT\}) = \tfrac{1}{4} = P(B_2)P(C_3)$$
$$P(A_1 \cap B_2 \cap C_3) = P(\emptyset) = 0 \neq \tfrac{1}{8} = P(A_1)P(B_2)P(C_3)$$

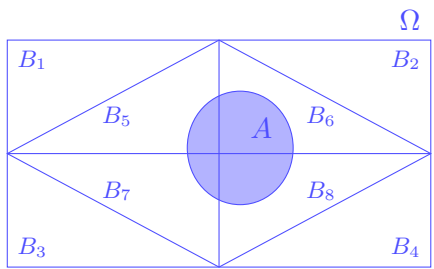
The Law of Total Probability

If *only the conditional probabilities and the probabilities of the conditional event are known*, the *total probability* of A results from

$$P(A) = P(A \mid B)P(B) + P(A \mid B^c)P(B^c).$$

Given a *partition of the sample space* Ω that is a sequence $B_1, B_2, B_3, \ldots$ of pair-wise disjoint events $B_i \cap B_j = \emptyset$ for all $i \neq j$ and $\bigcup_{i=1}^{\infty} B_i = \Omega$, then

$$P(A) = \sum_{i=1}^{\infty} P(A \cap B_i) = \sum_{i=1}^{\infty} P(A \mid B_i)P(B_i)$$



Bayes Theorem

Rewriting the definition of *conditional probability* for $P(A) > 0$ and $P(B) > 0$ gives the *Multiplication theorems*

$$P(A \cap B) = P(A \mid B)P(B)$$
$$P(A \cap B) = P(B \mid A)P(A)$$

which leads through equationg both to

BAYES THEOREM

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

This is very powerful and allows to "turn around" the conditioning.

An entire branch of statistics called Bayes Statistics is (basically) built on this formula (and the Bayesian interpretation of it).

Bayes Theorem

The *Bayes Theorem* describes the probability of an event based on *prior knowledge (before observation)* and allows to update the prior knowledge after observation to the *posterior (after observation)*.

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

Combining the Law of Total Probability given a partition $H_1, H_2, \ldots$ of Ω with Bayes Theorem, we get the alternative form

$$P(H_i \mid B) = \frac{P(B \mid H_i)P(H_i)}{P(B)} = \frac{P(B \mid H_i)P(H_i)}{\sum_{j=1}^{\infty} P(B \mid H_j)P(H_j)}$$

where the events H_i often referred to as *Hypothesis*. We call $P(H_i)$ *the prior* probability and $P(H_i \mid B)$ *the posterior* probability.

Bayes Theorem

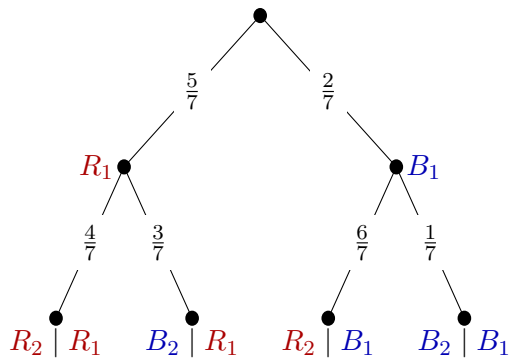
A urn contains *5 red* and *2 blue* marbles.

We draw a random marble and replace it with a marble of the other color, then a second marble is drawn.

1. What is the probability that the *second marble is red*?
2. What is the probability the *first marble was red given* the *second marble is red*?

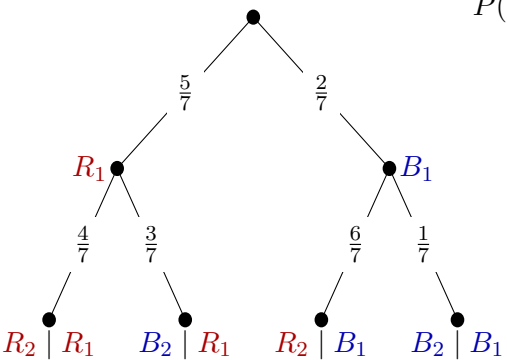
We use a *probability tree* to solve this problem;

Let R_i and B_i be the event of a *red* or *blue* marble on the i'th draw.



Bayes Theorem

Let R_i and B_i be the event of a *red* or *blue* marble on the i'th draw.



1. Using the *Law of Total Probability*

$$P(R_2) = P(R_2 \mid R_1)P(R_1) + P(R_2 \mid B_1)P(B_1)$$
$$= \frac{4}{7} \cdot \frac{5}{7} + \frac{6}{7} \cdot \frac{2}{7} = \frac{32}{49}$$

noting that $R_1 \cup B_1 = \Omega$.

2. Using *Bayes Theorem* we get

$$P(R_1 \mid R_2) = \frac{P(R_2 \mid R_1)P(R_1)}{P(R_2)}$$
$$= \frac{(4/7) \cdot (5/7)}{32/49} = \frac{5}{8}$$